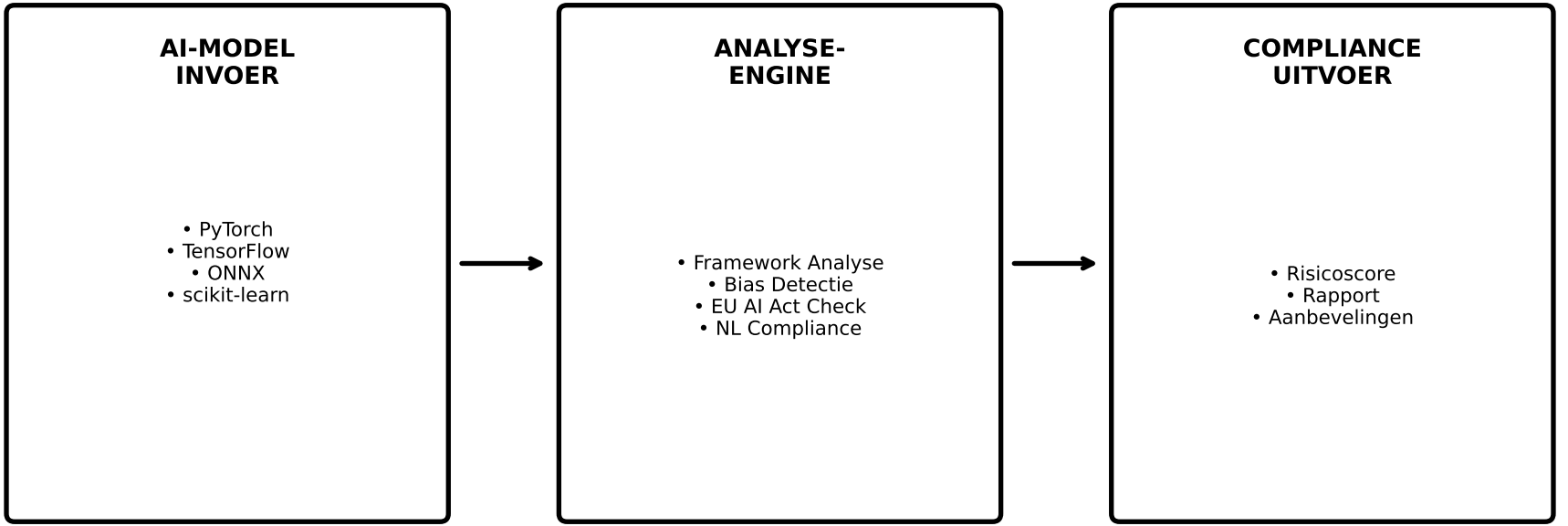
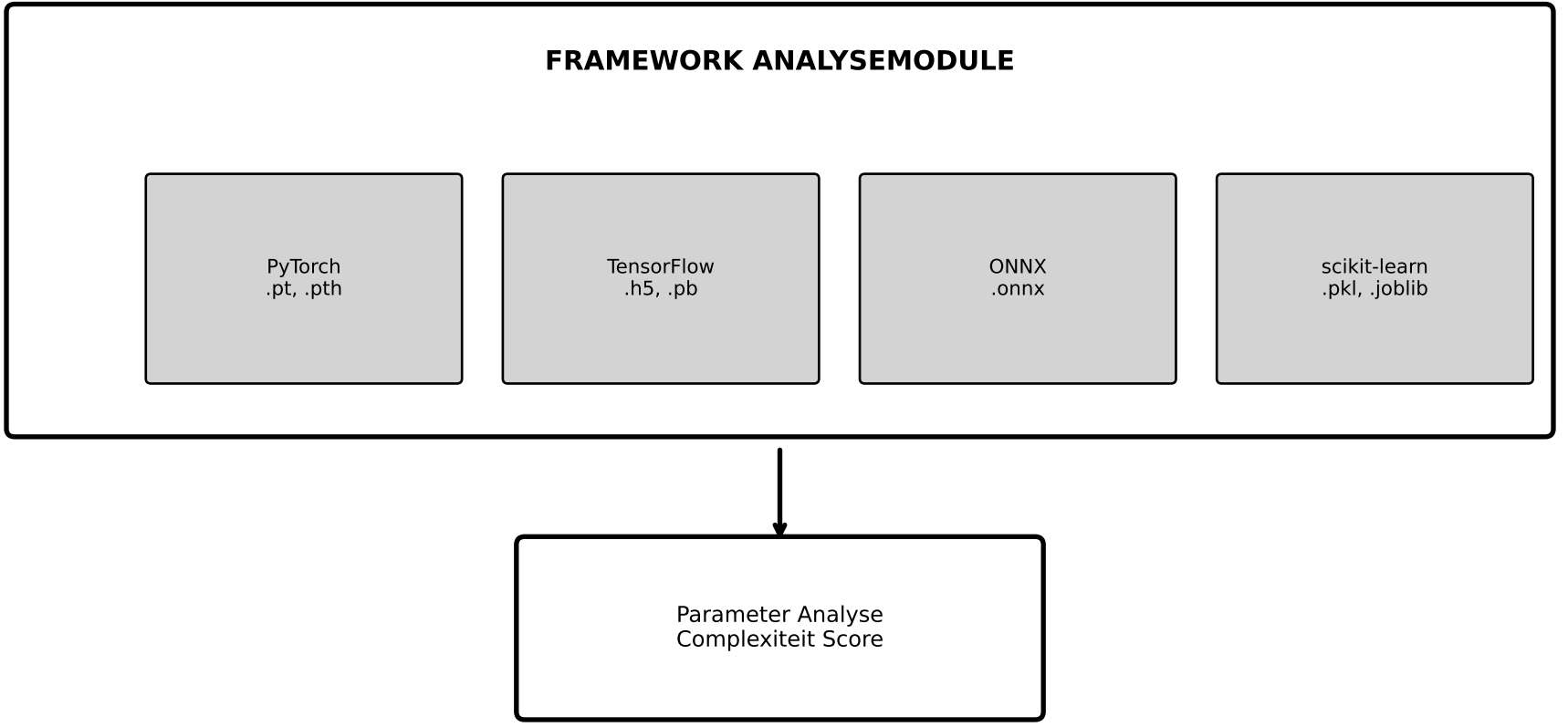
FIGUUR 1: Systeemarchitectuuroverzicht



FIGUUR 2: Multi-framework Analysemodule



FIGUUR 3: Bias-detectie-engine

BIAS-DETECTIE-ENGINE

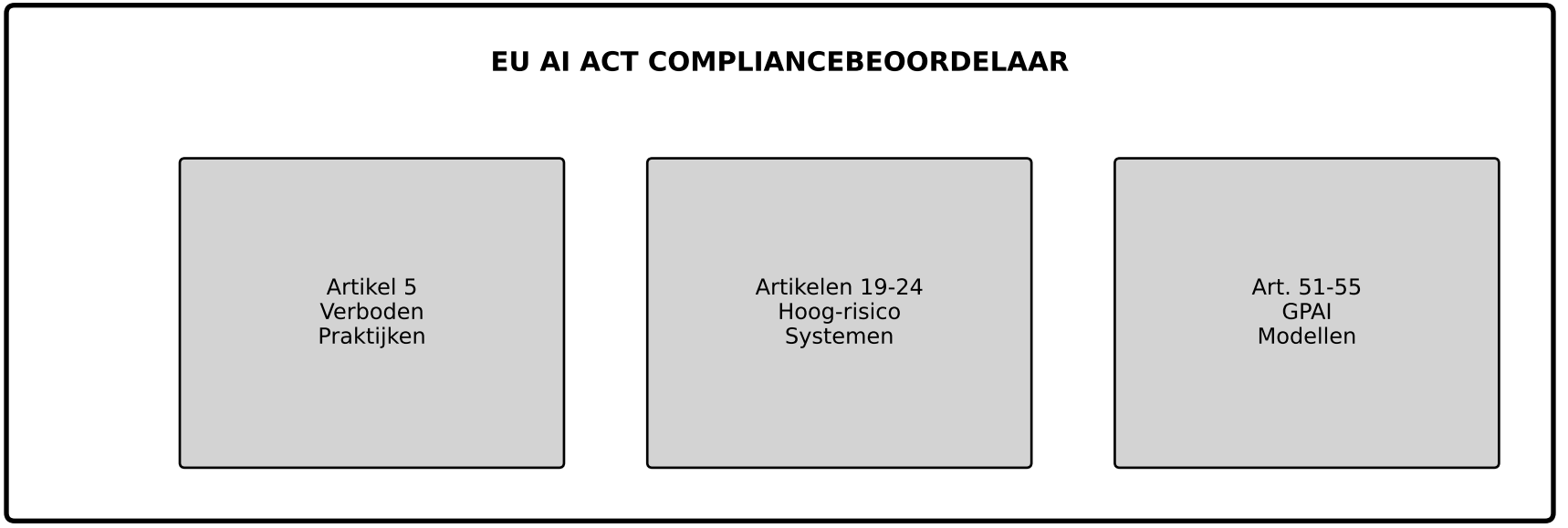
Demographic Parity: $P(Y=1|A=0) \approx P(Y=1|A=1)$

Equalized Odds: $TPR_0 \approx TPR_1$ en $FPR_0 \approx FPR_1$

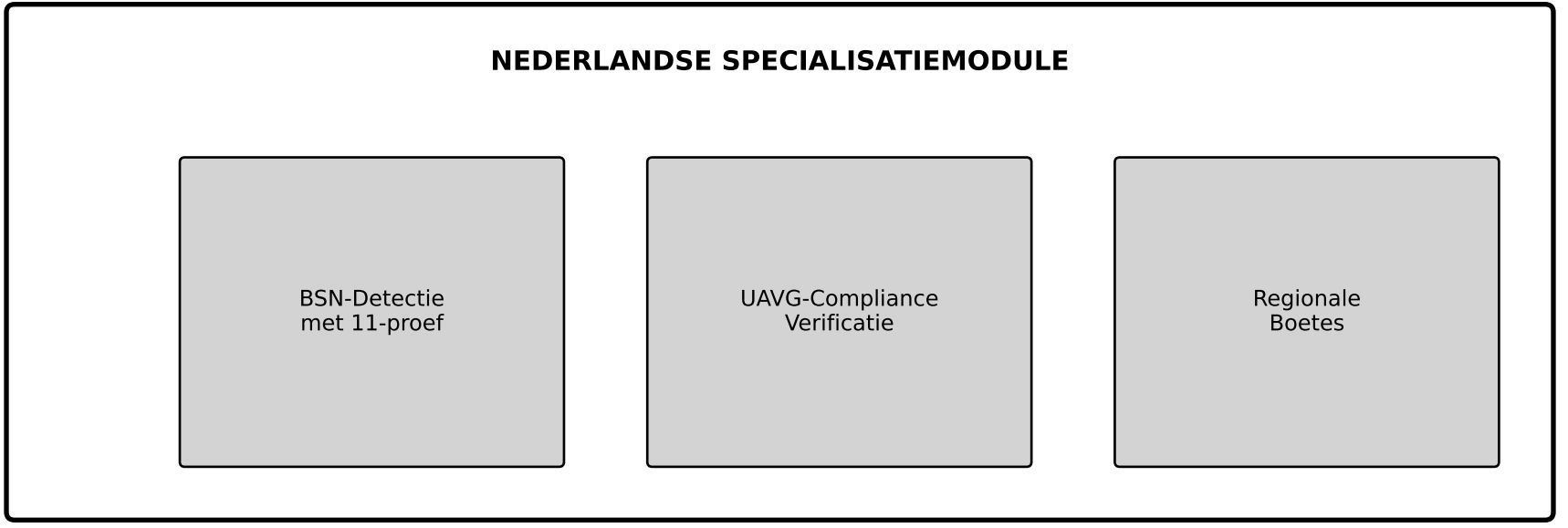
Calibration Score: $P(Y=1|Score=s,A=0) \approx P(Y=1|Score=s,A=1)$

Individual Fairness: $d(f(x_1),f(x_2)) \leq L \cdot d(x_1,x_2)$

FIGUUR 4: EU AI Act Compliancebeoordelaar



FIGUUR 5: Nederlandse Specialisatiemodule



FIGUUR 6: BSN Checksumvalidatie

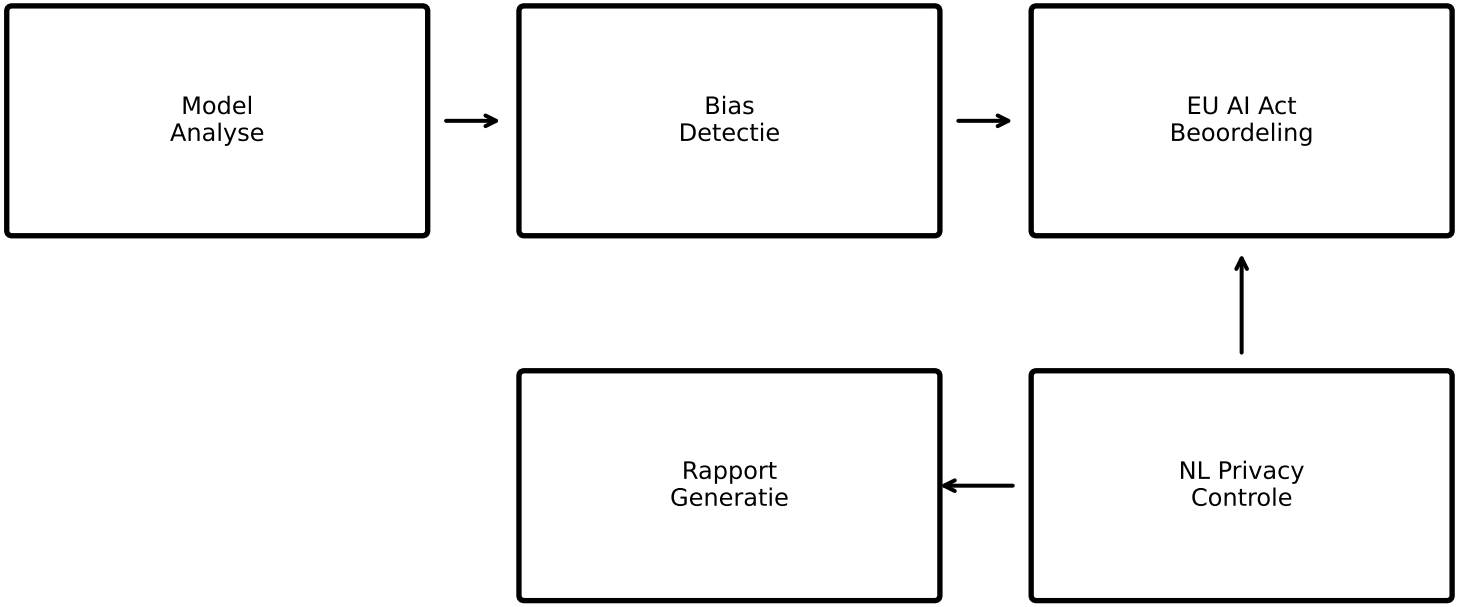
BSN CHECKSUMVALIDATIE (NEDERLANDSE 11-PROEF)

Formule:

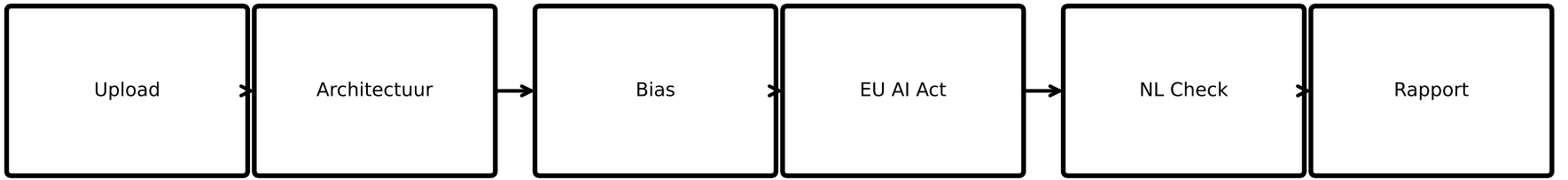
$$\text{checksum} = (d_0 \times 9) + (d_1 \times 8) + (d_2 \times 7) + (d_3 \times 6) + (d_4 \times 5) + (d_5 \times 4) + (d_6 \times 3) + (d_7 \times 2) - (d_8 \times 1)$$

Geldigheid: BSN is geldig indien checksum mod 11 = 0

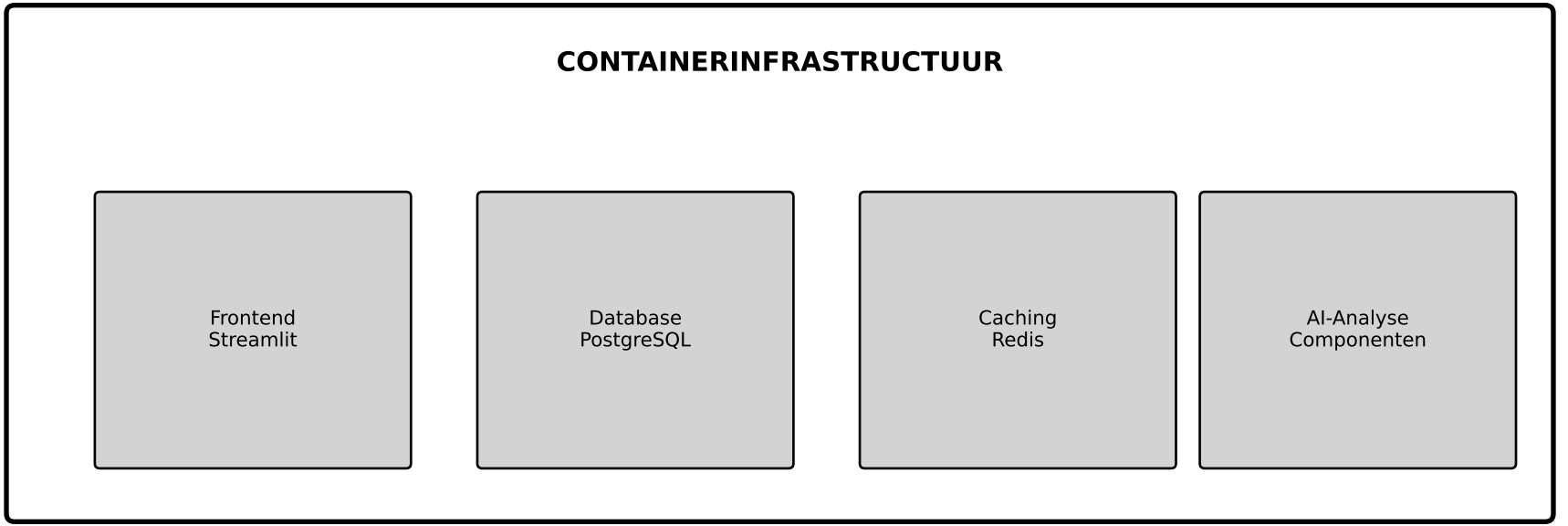
FIGUUR 7: Systeemflowdiagram



FIGUUR 8: Verwerkingspipeline



FIGUUR 9: Deploymentarchitectuur



FIGUUR 10: Vergelijkingsmatrix

Functie	Systeem	Handmatig	Concurrent
Multi-framework	Ja	Nee	Deels
Bias-detectie	Ja	Beperkt	Beperkt
EU AI Act 2025	Ja	Nee	Nee
Nederlandse module	Ja	Nee	Nee

FIGUUR 11: Waardepropositie

WAARDEPROPOSITIE

Kostenbesparing:	90% reductie vs handmatige audit
Risicoreductie:	Proactieve compliance
Snelheid:	<60 seconden per model
EU AI Act:	100% artikeldekking